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Skyborne Airline Academy

Power BI Dashboard - Project Methodology Summary

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PROJECT OVERVIEW

This dashboard was built in response to a case study prompt from Skyborne Airline Academy requesting a Power BI solution to centralize trainee and instructor performance reporting, and replace to manual spreadsheet-based tracking. The dataset provided contained records for 200 trainees across fields including airline placement, training duration, phase duration, student state of origin, and instructor assignment.

The dashboard was designed to serve three audiences simultaneously: executives requiring high-level KPI visibility, recruiters needing geographic and airline placement data, and training managers requiring instructor and cohort performance analysis.

DATA PREPARATION

Prior to analysis, I completed the following data preparation steps in Power Query:

- Removed 20 blank rows at the bottom of the dataset to prevent inflating or skewing any analyses or measure calculations
- Corrected column data types, including resolving a text-to-number conversion issue on the Average Time in Each Training Phase field
- Created a conditional column, Placement Status, using the logic: if Airline Partner is null then Not Placed, else Placed. This column served as the primary outcome variable throughout the dashboard
- Created a calculated column, Training Duration Group, using nested IF statements to bucket trainees into five duration ranges: 11-14, 15-18, 19-22, 23-26, and 27+ months. This enabled distribution analysis and cohort comparisons that would not have been readable at the individual month level

DAX MEASURES AND MODELING

All DAX measures were stored in a dedicated _Measures table, hidden from Report View to maintain a clean field list. Sixteen measures were created covering placement rates, attrition, training duration averages, airline partner metrics, and instructor performance statistics including a rate gap measure quantifying the performance spread between the highest and lowest performing instructors.

Key modeling decisions included using DIVIDE() for all rate calculations to handle zero denominators and using contextual patterns with CALCULATE() to enable measures like Placement Rate by Instructor and Placement Rate by State to automatically filter to the relevant context when used in visuals.

DASHBOARD STRUCTURE

The dashboard is organized across five pages:

- **Page 1: Executive Overview:** Five KPI cards providing an at-a-glance summary of total trainees, placement rate, attrition rate, and average training and phase durations. Supported by a donut chart showing placed vs. unplaced distribution and a clustered bar chart showing raw placement counts by instructor.
- **Page 2: Training Performance:** Training duration distribution across five cohort buckets, a table of extended training outliers exceeding 27 months, and a clustered bar chart showing placement outcomes by training duration group. These were annotated to underscore the key finding that training duration does not predict placement outcome.
- **Page 3: Airline Placement:** Two KPI cards covering total airline partners and average placements per airline, a filled map showing geographic placement performance by state using a gradient color scale, and a horizontal bar chart showing placement rates by state ranked in descending order. This was annotated to flag Indiana and Missouri as underperforming markets and gray states as untapped recruitment opportunities.

- **Page 4: Instructor Performance Analysis:** Two KPI cards showing top instructor rate and the instructor rate gap, supported by an Instructor Performance Summary matrix with color-coded placement rates. Annotated to contextualize the 28.7 percentage point performance gap between the strongest and weakest instructors.
- **Page 5: Insights, Recommendations and Limitations:** A structured two-column summary page presenting key analytical findings and actionable recommendations alongside an honest accounting of data limitations that precluded full coverage of all stated business objectives.

KEY ANALYTICAL FINDINGS

- Skyborne achieved an 83.5% overall placement rate across 25 airline partners, placing 167 of 200 trainees
- Instructor assignment is the strongest predictor of placement success. Charlotte Brooks and Mia Collins achieved 96.6% placement rates while Daniel Cooper achieved 67.9%, a 28.7 percentage point gap
- Training duration does not predict placement outcome. Placed and unplaced trainees averaged 18.0 and 18.9 months respectively, a difference too small to be actionable
- Arizona, Colorado, Illinois, Louisiana, New York, Ohio, Tennessee, Virginia, and Washington achieved near-perfect placement rates and represent Skyborne's strongest recruitment markets
- Indiana and Missouri consistently underperform and warrant targeted employer outreach and network development
- Placements are evenly distributed across all 25 airline partners at approximately 6.7 trainees per airline, indicating no single-airline dependency risk

DATA LIMITATIONS ACKNOWLEDGED

The following business objectives could not be fully addressed due to limitations in the available dataset:

- **Per-phase training duration:** Only a single average phase duration per trainee was available. True phase-by-phase breakdown was not possible. Recommendation: Capture individual phase completion dates in the source system
- **Phase-level dropout analysis:** Only final placement outcome was recorded with no milestone or checkpoint data. Recommendation: Add phase completion status fields to enable real-time dropout tracking
- **Time to placement after graduation:** No graduation date or hire date fields exist in the dataset. Recommendation: Add graduation and first employment dates to enable this metric in future iterations
- **Demographic cohort analysis:** No demographic fields such as age, educational background, or prior aviation experience were available. Cohort analysis was limited to geography, training duration, and instructor assignment. Recommendation: Collect relevant background data at enrollment to enable richer segmentation

DASHBOARD DESIGN RATIONALE

The dashboard was built to match Skyborne Airline Academy's brand identity. The canvas background color (#001125) was matched to Skyborne's website navy and the Skyborne logo was applied consistently across all five pages. A turquoise (#00B4C8) and amber (#F0A500) color convention was applied throughout, with turquoise representing positive outcomes (placed trainees, high performers) and amber representing attrition or underperformance. Annotation text boxes were added across Pages 2, 3, and 4 to surface key insights directly within the relevant visual context, reducing the interpretive burden on the viewer and ensuring the dashboard communicates findings rather than simply displaying data.